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United States Patent Application Publication

20250273787

Kind Code

A1

Publication Date

August 28, 2025

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BATTERY PACK

Abstract

The battery pack comprises a plurality of cell modules; and at least one battery-related device, wherein the plurality of cell modules are arranged side by side in a longitudinal direction thereof, with an intermediate space interposed therebetween, and the at least one battery-related device is disposed inside the intermediate space or above the intermediate space.

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Family ID: 1000008489020

Appl. No.: 19/062012

Filed: February 25, 2025

Foreign Application Priority Data

JP	2024-028804	Feb. 28, 2024
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Publication Classification

Int. Cl.: H01M50/204 (20210101); H01M10/613 (20140101); H01M10/6563 (20140101); H01M50/583 (20210101)

U.S. Cl.:

CPC H01M50/204 (20210101); H01M10/613 (20150401); H01M10/6563 (20150401); H01M50/583 (20210101); H01M2200/103 (20130101); H01M2220/20 (20130101)

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-028804 filed on Feb. 28, 2024, which is incorporated herein by reference in its entirety including the specification, claims, drawings, and abstract.

TECHNICAL FIELD

[0002] This specification discloses a battery pack mounted on a vehicle.

BACKGROUND

[0003] A battery pack for charging and discharging electric power is mounted on an electric vehicle using a motor as a power source. The battery pack typically includes a plurality of cell modules and one or more battery-related devices. The cell module is a component in which a plurality of battery cells are formed into one component. The battery-related device is, for example, an ECU, a connector, a duct, or the like.

[0004] Patent Document 1 discloses a battery pack in which a plurality of cell modules are arranged in the vehicle longitudinal direction. In the Patent Document 1, the cell module has a substantially rectangular parallelepiped shape elongated in the vehicle width direction. In the Patent Document 1, the battery-related device is disposed adjacent to the cell module in the vehicle width direction.

[0005] Here, in order to reduce the dimension of the battery pack in the short direction, a plurality of cell modules may be arranged in the longitudinal direction of the cell modules. In this case, there is a problem that the longitudinal dimension of the battery pack becomes excessively large depending on the arrangement of the battery-related devices.

[0006] Accordingly, the present specification discloses a battery pack in which the arrangement efficiency of parts is improved.

CITATION LIST

[0007] PATENT DOCUMENT 1: JP.2008-234870.A

SUMMARY

[0008] A battery pack disclosed in the present specification comprises: a plurality of cell modules; and at least one battery-related device, wherein the plurality of cell modules are arranged side by side in a longitudinal direction thereof, with an intermediate space interposed therebetween, and the at least one battery-related device is disposed inside the intermediate space or above the intermediate space.

[0009] In this case, the at least one battery-related device may include: an electronic device disposed above the intermediate space; and a connector that is disposed below the electronic device and electrically relays the electronic device and a wire harness, and a dimension of the intermediate space in the longitudinal direction may be smaller than a dimension of the electronic device in the longitudinal direction.

[0010] The at least one battery-related device may include: a fuse disposed inside the intermediate space; and a connector which is disposed inside the intermediate space or above the intermediate space and to which a wire harness extending from each of the plurality of cell modules may be connected.

[0011] Further, the connector may be disposed in an orientation such that an insertion opening of the wire harness faces downward.

[0012] The at least one battery-related device may include a duct, a part of which is disposed inside the intermediate space.

[0013] According to the technology disclosed in the present specification, the arrangement efficiency of parts is improved.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0014] Embodiment(s) of the present disclosure will be described based on the following figures, wherein:

[0015] FIG. 1 is a perspective view of a battery pack;

[0016] FIG. 2 is a schematic view showing an arrangement of battery packs;

[0017] FIG. 3 is a schematic diagram showing a battery controller and an arrangement of the battery controller;

[0018] FIG. 4 is a schematic diagram showing the arrangement of fuses;

[0019] FIG. 5 is a schematic view showing an arrangement of ducts; and

[0020] FIG. 6 is a diagram showing an example of another battery pack.

DESCRIPTION OF EMBODIMENT

[0021] Hereinafter, a configuration of the battery pack **10** will be described with reference to the drawings. FIG. 1 is a schematic perspective view of a battery pack **10**. In the drawings, Fr, Up, and Rh indicate the front side, the upper side, and the right side of the vehicle, respectively.

[0022] The battery pack **10** shown in FIG. 1 is mounted on an electric vehicle and supplies electric power to a traveling motor. The electric vehicle is a vehicle having a motor as one of power sources, and corresponds to, for example, a hybrid electric vehicle, a battery electric vehicle, a fuel cell electric vehicle, and the like. The battery pack **10** is disposed in a place that is not in the way of an occupant, for example, below a luggage space or an in-vehicle seat. In this example, the battery pack **10** is disposed below the rear seat **100**, as shown in FIG. 2. In addition, in the present example, the battery pack **10** is disposed in a posture in which the longitudinal direction of a cell module **12**, which will be described later, is parallel to the vehicle width direction. Therefore, the arrow Rh indicating the right side in FIGS. 1 to 6 is the same direction as the longitudinal direction of the cell module **12**, and the arrow Fr indicating the front side is the same direction as the short direction of the cell module **12**.

[0023] The battery pack **10** is configured by housing a plurality of cell modules **12** and one or more battery-related devices in one case (not shown). The cell module **12** is configured by alternately stacking a plurality of battery cells **14** and a plurality of separators **15** in the thickness direction. The battery cell **14** is a secondary battery that can be charged and discharged. For example, the battery cell **14** is a lithium ion secondary battery or a sodium ion secondary battery. The battery cell **14** may be an all-solid-state battery. The battery cell **14** is a rectangular battery having a flat rectangular parallelepiped shape. The plurality of battery cells **14** are electrically connected in series or in parallel by conductors called bus bars. The separator **15** is a plate made of an insulating material. Grooves that function as flow paths through which the cooling fluid flows are formed on the surface of the separator **15**.

[0024] The cell module **12** in which the plurality of battery cells **14** and the plurality of separators **15** are stacked has a substantially rectangular parallelepiped shape elongated in the stacking direction. The battery pack **10** includes two cell modules **12**. In the present example, the two cell modules **12** are arranged side by side in the longitudinal direction of the cell module **12**. The reason for this arrangement is to improve space efficiency, which will be described later. The battery pack **10** is disposed such that the longitudinal direction of the cell module **12** is substantially parallel to the vehicle width direction. Therefore, in the present example, the longitudinal direction of the cell module **12** is substantially parallel to the vehicle width direction, and the short direction of the cell module **12** is substantially parallel to the vehicle longitudinal direction.

[0025] An intermediate space **16** is provided between two adjacent cell modules **12**. The size of the intermediate space **16** is not particularly limited. However, if the intermediate space **16** is excessively large, the dimension of the battery pack **10** in the longitudinal direction becomes large. On the other hand, if the intermediate space **16** is excessively small, the battery-related device to be described later may not be disposed, or the workability of the connection work of the wire harness **22** may be deteriorated. Therefore, the intermediate space **16** may have, for example, a size that allows the hand of the operator to enter. Therefore, the intermediate space **16** may be set in a range of 5 cm or more and less than 20 cm, for example.

[0026] The battery pack **10** further includes one or more battery-related devices. The battery-related device is a device necessary for safely driving the battery pack **10**. For example, the battery controller **18**, the connector **20**, the fuse **24**, the duct **25**, the junction box **30**, and the service plug **32** correspond to the battery-related device. The battery controller **18** is a computer that controls charging and discharging of the cell module **12**, and is generally an electronic device called an ECU (Electronic Control Unit). The battery controller **18** has a flat rectangular shape and is disposed above the intermediate space **16**.

[0027] A wire harness **22** drawn from the cell module **12** is electrically connected to the connector **20**. For example, the connector **20** is a female connector into which a male connector attached to the end of the wire harness **22** is inserted. The connector **20** is also electrically connected to the battery controller **18**. The connector **20** relays power or electrical signals between the battery controller **18** and the cell module **12**. As shown in FIG. 1, the connector **20** is disposed below the battery controller **18** in a posture in which the insertion port of the wire harness **22** faces downward.

[0028] The fuse **24** is an electronic component that protects the cell module **12** from overcurrent. As shown in FIG. 1, the fuse **24** is disposed inside the intermediate space **16**. The duct **25** is a flow path that allows the cell module **12** to communicate with a cooling flow path and a blower (not shown). As shown in FIG. 1, at least a part of the duct **25** is disposed inside the intermediate space **16**. The duct **25** and the fuse **24** are arranged in the short direction of the cell module **12** in the intermediate space **16**.

[0029] The junction box **30** is an electronic component in which relays for allowing or blocking the flow of electric power are unitized. The junction box **30** is disposed above the cell module **12**. The service plug **32** is a component provided at an intermediate point of the power supply circuit, and is a component that cuts off a high voltage by being removed. When servicing the battery pack **10**, the operator removes the service plug **32** in advance. After the maintenance is completed, the operator reattaches the service plug **32**. In this example, the service plug **32** is disposed above the cell module **12** and next to the junction box **30**.

[0030] As is clear from the above description, in the present example, the two cell modules **12** are arranged side by side in the longitudinal direction of the cell modules **12**. The reason for this configuration will be described. As described above, in the present embodiment, the battery pack **10** is disposed below the rear seat **100**. The dimension in the front-rear direction of the lower side of the rear seat **100** is smaller than the dimension in the vehicle width direction. Therefore, the size of the battery pack **10** disposed on the lower side of the rear seat **100** in the front-rear direction is required to be as small as possible.

[0031] Therefore, in this example, the two cell modules **12** are arranged in the longitudinal direction, and the battery pack **10** is disposed below the rear seat **100** in a posture in which the longitudinal direction is substantially parallel to the vehicle width direction. With this arrangement, the dimension of the battery pack **10** in the vehicle front-rear direction can be suppressed to be small.

[0032] Further, in the present embodiment, the intermediate space **16** is provided between the two cell modules **12**, and the battery-related device is provided inside or above the intermediate space **16**. The reason for this arrangement will be described with reference to FIGS. 3 to 5. FIG. 3 is a

schematic front view of the battery pack **10**. The state **S1** in the upper part of FIG. **3** shows the arrangement of the present example, and the state **S2** in the middle part of FIG. **3** and the state **S3** in the lower part of FIG. **3** show the arrangement of the comparative example, respectively.

[0033] First, as in the state **S2**, a case where the two cell modules **12** are arranged without a gap and the battery controller **18** and the connector **20** are disposed outside in the longitudinal direction is considered. In this case, the dimension of the battery pack **10** in the longitudinal direction is increased by the width of the battery controller **18**. Further, in the case of the state **S2**, since the distance from the cell module **12** on the right side of the drawing to the connector **20** becomes long, the wire harness **22** becomes long. Further, in the case of the state **S2**, the battery controller **18** and the connector **20** are disposed within the range in the height direction of the cell module **12**. In this case, it is difficult to secure a sufficient work space below the connector **20**. As a result, the workability of the connection of the wire harness **22** is poor. If the insertion opening of the connector **20** is directed upward, the workability is improved, but in this case, another problem is caused in that foreign matter easily adheres to the insertion opening.

[0034] Therefore, as shown in the state **S3**, two cell modules **12** may be disposed without a gap therebetween, and the battery controller **18** and the connector **20** may be provided above the cell modules **12**. With this arrangement, the dimension of the battery pack **10** in the longitudinal direction can be reduced. However, in this case, in order to secure a work space for connecting the wire harness **22** to the connector **20**, it is necessary to dispose the battery controller **18** and the connector **20** so as to be largely separated from the cell module **12**. As a result, in the state **S3**, the dimension of the battery pack **10** in the height direction becomes significantly large, and it becomes difficult to dispose the battery pack **10** below the rear seat **100**.

[0035] On the other hand, in this example, as shown in the state **S1**, the intermediate space **16** is provided between the two cell modules **12**. The battery controller **18** and the connector **20** are disposed above the intermediate space **16**. In this case, the dimension of the battery pack **10** in the longitudinal direction can be made smaller than that in the state **S2**. In addition, the wire length of the wire harness **22** can be significantly reduced. Further, in this case, the intermediate space **16** can be used as a work space for connection of the wire harness **22**. As a result, in the state **S1**, the battery controller **18** and the wire harness **22** can be disposed near the cell module **12**. Accordingly, the size of the battery pack **10** in the height direction can be suppressed to be small.

[0036] In this example, in order to sufficiently reduce the dimension of the entire battery pack **10** in the longitudinal direction, the dimension of the intermediate space **16** in the longitudinal direction is made smaller than either the dimension of the battery controller **18** in the longitudinal direction or the dimension of the connector **20** in the longitudinal direction. However, when the installation space of the battery pack **10** is sufficiently wide, the dimension of the intermediate space **16** in the longitudinal direction may be larger than the dimension of the battery controller **18** in the longitudinal direction or the dimension of the connector **20** in the longitudinal direction. Both the battery controller **18** and the connector **20** or only the connector **20** may be disposed inside the intermediate space **16**. With this configuration, the size of the battery pack **10** in the height direction can be suppressed to be small. In addition, the connector **20** may be disposed in a posture in which the insertion port faces upward as long as a sufficient dustproof measure can be taken.

[0037] Next, the arrangement of the fuses **24** will be described with reference to FIG. **4**. FIG. **4** is a schematic front view of the battery pack **10**. A state **S4**, which is the upper stage of FIG. **4**, shows the arrangement of the present example, and a state **S5**, which is the lower stage of FIG. **4**, shows the arrangement of the comparative example. When the two cell modules **12** are disposed without a gap and the fuse **24** is disposed outside in the longitudinal direction as in the state **S5**, the dimension of the battery pack **10** in the longitudinal direction becomes large, and the wire length of the wire harness **22** also becomes long. When the fuse **24** is disposed above the cell module **12**, the height of the battery pack **10** increases.

[0038] On the other hand, in this example, as shown in the state **S4**, the fuse **24** is disposed inside

the intermediate space **16**. As a result, the wire length of the wire harness **22** is reduced. In this example, the dimension of the fuse **24** in the longitudinal direction is smaller than either the dimension of the connector **20** in the longitudinal direction or the dimension of the intermediate space **16** in the longitudinal direction. However, the arrangement and size of the fuses **24** are merely examples, and the fuses **24** may be disposed outside the intermediate space **16**. For example, the fuse **24** may be disposed on the intermediate space **16** or on the cell module **12**. [0039] Next, the arrangement of the duct **25** will be described with reference to FIG. 5. FIG. 5 is a schematic top view of the battery pack **10**. In FIG. 5, a state **S6** in the upper stage shows the arrangement of the present example, and a state **S7** in the lower stage shows the arrangement of the comparative example. As shown in FIG. 5, the duct **25** is roughly divided into a sub duct **27** connected to each of the cell modules **12**, and a main duct **26** in which a plurality of sub ducts **27** merge.

[0040] When the two cell modules **12** are disposed without a gap as in the state **S7**, the main duct **26** needs to be disposed on the front side or the rear side of the cell modules **12**. For example, a case where the main duct **26** is disposed on the front side of the two cell modules **12** is considered. In this case, the sub duct **27** connecting the main duct **26** and the cell module **12** is connected to the main duct **26** through the front side of the cell module **12** from the outside in the longitudinal direction of the cell module **12**. As a result, in this case, the path of the sub duct **27** becomes long, and a large curve is generated in the middle of the path of the sub duct **27**. As a result, the pressure loss increases, and thus the cooling efficiency of the cell module **12** decreases. In addition, in the case of the state **S7**, since a space for arranging the sub duct **27** is necessary, the dimension of the battery pack **10** in the short direction increases accordingly.

[0041] On the other hand, in this example, as in the state **S6**, the intermediate space **16** is provided between the two cell modules **12**, and a part of the main duct **26** is disposed inside the intermediate space **16**. In this case, the sub duct **27** extends linearly from the longitudinally inner end of the cell module **12** toward the main duct **26**. As is clear from FIG. 5, in the case of the state **S6**, the sub duct **27** is significantly shorter than in the case of the state **S7**, and the curve of the sub duct **27** is small. As a result, in the state **S6**, since the pressure loss can be reduced, the cooling efficiency of the cell module **12** is improved. Further, in the case of the state **S6**, since the sub duct **27** can be disposed within the range in the lateral direction of the cell module **12**, the dimension of the battery pack **10** in the lateral direction is reduced.

[0042] The arrangement of the duct **25** is also an example, and the duct **25** may be arranged outside the intermediate space **16**. For example, the duct **25** may be disposed outside in the longitudinal direction or outside in the lateral direction of the two cell modules **12**.

[0043] In the above description, the battery pack **10** includes the two cell modules **12**. However, the number of the cell modules **12** is not particularly limited as long as it is two or more, and may be, for example, three. In this case, as shown in the upper part (state **S8**) of FIG. 6, the battery pack **10** may have two intermediate spaces **16**. A part of the duct **25** may be disposed in each of the two intermediate spaces **16**.

[0044] As long as the battery pack **10** has a plurality of cell modules **12** adjacent to each other in the longitudinal direction, the battery pack may further have cell modules **12** adjacent to each other in the short direction. That is, as shown in the lower part (state **S9**) of FIG. 6, four or more cell modules **12** may be arranged in an array in the longitudinal direction and the short direction.

[0045] Further, in the above description, the battery pack **10** is disposed in a posture in which the longitudinal direction of the cell module **12** is substantially parallel to the vehicle width direction of the vehicle. However, the arrangement and orientation of the battery pack **10** in the vehicle may be changed as appropriate. For example, the battery pack **10** may be disposed such that the longitudinal direction of the cell module **12** is substantially parallel to the longitudinal direction of the vehicle.

REFERENCE SIGNS LIST

[0046] **10** battery pack, **12** cell module, **14** battery cell, **15** separator, **16** intermediate space, **18** battery controller, **20** connector, **22** wire harness, **24** fuse, **25** duct, **26** main duct, **27** sub duct, **30** junction box, **32** service plug, **100** rear seat.

Claims

1. A battery pack comprising: a plurality of cell modules; and at least one battery-related device, wherein the plurality of cell modules are arranged side by side in a longitudinal direction thereof, with an intermediate space interposed therebetween, and the at least one battery-related device is disposed inside the intermediate space or above the intermediate space.
 2. The battery pack according to claim 1, wherein the at least one battery-related device includes: an electronic device disposed above the intermediate space; and a connector that is disposed below the electronic device and electrically relays the electronic device and a wire harness, and a dimension of the intermediate space in the longitudinal direction is smaller than a dimension of the electronic device in the longitudinal direction.
 3. The battery pack according to claim 1, wherein the at least one battery-related device includes: a fuse disposed inside the intermediate space; and a connector which is disposed inside the intermediate space or above the intermediate space and to which a wire harness extending from each of the plurality of cell modules is connected.
 4. The battery pack according to claim 2, wherein the connector is disposed in an orientation such that an insertion opening of the wire harness faces downward.
 5. The battery pack of claim 1, wherein the at least one battery-related device includes a duct, a part of which is disposed inside the intermediate space.
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